

Ankit

Master of Computer Applications
in Artificial Intelligence and Machine Learning
Chandigarh University, Punjab

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Summary

Software Engineer with expertise in AI, Machine Learning, Deep Learning, Full-Stack and Mobile development. Experienced in real-time streaming pipelines, NLP systems, fraud detection, and cloud-backed enterprise applications. Led a team at CSIR-NPL to build an employee management system.

Experience

CSIR – National Physical Laboratory (NPL)

April 2024 – May 2024

Software Engineering Intern

- Built a secure digital employee ID system using Firebase Authentication and role-based access control (RBAC), serving 200+ staff.
- Implemented QR-code-based selective data sharing with Firebase Firestore and Firestorage for real-time synchronization across departments.
- Developed internal workflow features (restroom booking, canteen ordering) using cloud-backed serverless architecture.

Projects

Federal.AI: Privacy-Preserving Federated Learning Platform [Github](#)

Nov 2025 – Dec 2025

Architected a privacy-preserving federated learning system for ICU biomarker discovery using MIMIC-IV, enabling decentralized model training across hospitals without sharing sensitive patient data.

- **Tools & technologies:** PyTorch, FastAPI, React.js, Docker, Federated Learning, Differential Privacy, Secure Aggregation, SQLite (SQLModel)
- Engineered federated server–client architecture simulating multi-hospital training with encrypted gradient sharing and secure aggregation protocols.
- Designed FL orchestration pipeline with round-based training, global model aggregation, and local hospital training workflows in PyTorch.
- Built full-stack monitoring dashboard for experiment tracking, hospital management, and compliance visualization using React and FastAPI.
- Achieved high-performance biomarker discovery with strong generalization (up to ~0.93 AUC) across federated ICU datasets.

Diffusion Fine-Tuning Platform: LoRA-Based Parameter-Efficient Model Adaptation [Github](#) · [Vercel](#)

Feb 2025 – Apr 2025

Full-stack platform for parameter-efficient fine-tuning (LoRA) and deployment of Stable Diffusion models on custom datasets, combining ML research with production-ready architecture.

- **Tools & technologies:** PyTorch, HuggingFace Diffusers, PEFT, FastAPI, Next.js, Supabase, Docker, CLIP
- Applied LoRA to Stable Diffusion UNet attention layers, cutting trainable parameters to ~0.5–2% of total — enabling fine-tuning on consumer GPUs with 8 GB VRAM.
- Built adaptive training pipeline with real-time SSE loss streaming, EMA smoothing, and gradient norm tracking for live training visibility.
- Deployed via Docker Compose with FastAPI backend and Next.js frontend; integrated dual-layer NSFW filtering, CLIP score evaluation, and cross-platform MPS/CUDA support.

eyeBERT – Bilingual Medical Chatbot

Oct 2024

Pioneered eyeBERT, a BERT-based AI assistant that streamlines hospital onboarding by answering FAQs on policies, procedures, and departmental guidelines in both Tamil and English.

- **Tools & technologies:** BERT, Hugging Face Transformers, Streamlit, ngrok, NLP, Translation Models
- Fine-tuned a question-answering BERT model on a custom organizational and medical dataset to answer 50+ FAQs on policies, case feedback, and departmental procedures.
- Enabled bilingual support by integrating a Tamil translation toggle for seamless interaction in 2 languages without degrading model accuracy.
- Configured a Streamlit + ngrok interface for real-time onboarding assistance, achieving 85%+ answer accuracy on internal validation with positive pilot.

Education

Degree/Certification	Institute/Board	CGPA/Percentage	Year
MCA (AI ML)	Chandigarh University	6.73	2024 - 26
BCA	K.R. Mangalam University	7.53	2021 - 24
Senior Secondary	CBSE	73%	2019 - 20

Technical Skills

Programming: Python, SQL, JavaScript, Typescript

AI / ML: PyTorch, TensorFlow, Scikit-Learn, CatBoost, Hugging Face Transformers, BERT, LLM Agents, RAG, LangChain, Prompt Engineering, Vector Databases, Generative AI, NLP, AI Evaluation, Multi-Agent Systems, Fine-Tuning, LoRA, PEFT, CLIP

Backend & Infrastructure: FastAPI, Docker, Kafka, PostgreSQL, Firebase, Supabase, REST APIs, Git, Postman

Frontend & AI Interfaces: React.js, Next.js, Conversational AI Interfaces, Streaming Responses, API Integration, Real-Time AI Workflows

Certification

IBM – Big Data 101 | Samatrix – Data Science & Data Analysis, Data Analysis Using Python, Foundation to AI Data Science | Postman Academy – API Fundamentals Student Expert | Amazon – Full Stack Development

Research Work

- Privacy-Preserving Federated Learning for Biomarker Discovery in ICU Systems (Manuscript under submission) [Project Link](#)
- LoRA Fine-Tuning Experiments for Stable Diffusion Models (Manuscript under submission) [Project Link](#)